

Candidate Name: _____

Class: 07S _____



JURONG JUNIOR COLLEGE

J2 Preliminary Exam 2008

H2 Biology 9747/3

THURSDAY

21 AUGUST 2008

1 HOUR 30 MIN

INSTRUCTIONS TO CANDIDATES:

Write your name and class in the spaces at the top of this page and on any separate answer paper used.

Answer **all** questions.

At the end of the examination

1. fasten any separate answer paper used securely to the question paper.

EXAMINER'S USE ONLY	
1	
2	
3	
4	
Total	

INFORMATION FOR CANDIDATES:

The intended number of marks is given in brackets [] at the end of each question or part question.

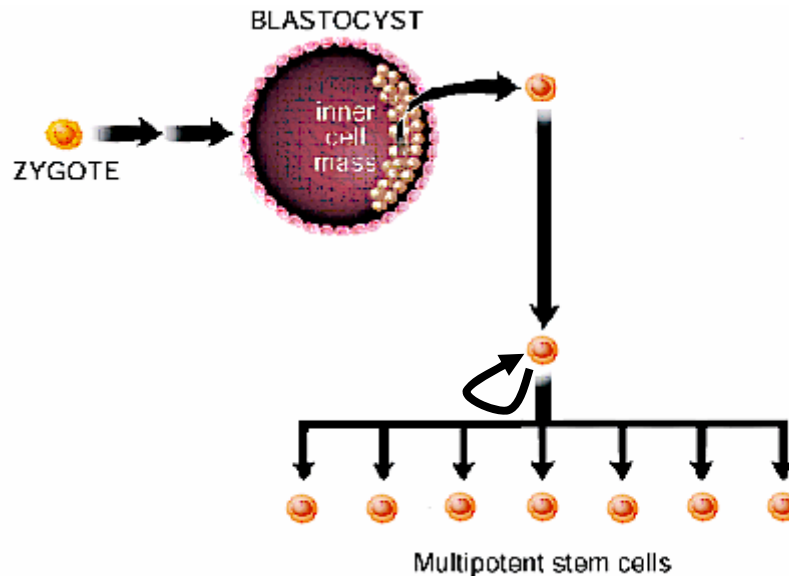
This Paper consists of **8** printed pages.

[Turn over]

Section A

Answer **all** questions.

- The figure below shows how a one-celled zygote undergoes various cellular divisions to give rise to other cell types.



- Describe two unique features of stem cells that could be seen from the figure above. [2]

- Stem cells can undergo differentiation to give rise to specialized cell types.;;
- Stem cells are capable of dividing indefinitely and producing copies of themselves / capable of self renewal.;;

- Describe the developmental potential of the one-celled zygote and the inner cell mass of the blastocyst. [4]

- One celled zygote is a totipotent stem cell;;
- that can form extraembryonic membranes and the embryo;;
- Inner cell mass of the blastocyst are all pluripotent stem cells;;
- that can form all cell types of a human.;;

- Multipotent stem cells can form specialized cell types of the tissue in which they reside. Give an example of a multipotent stem cell and the specialized cell types that it can form. [2]

- Hematopoietic stem cells.;;
- They can give rise to red blood cells and white blood cells.;;

OR

- Bone marrow stromal stem cells.;;
- They can give rise to bone cells/osteocytes, cartilage cells/chondrocytes, fat cells/adipocytes and other kinds of connective tissue cells such as those in tendons.;;

OR

- **Neural stem cells.;;**
- **They can give rise to neurons and non-neuronal cells – astrocytes and oligodendrocytes.;;**

(d) Adult stem cells are multipotent stem cells found among differentiated cells in a tissue or organ. What are the primary roles of these adult stem cells and describe one appropriate unique feature of this type of stem cells that is not mentioned in (a). [2]

- **Their primary role is to maintain and repair the specific tissues in which they reside.;;**
- **They can remain undifferentiated / unspecialized for long periods of time.;;**

Parkinson's disease is a disorder of the central nervous system involving the brain and spinal cord. Parkinson's disease occurs when certain nerve cells in a part of the brain become impaired. A chemical substance called dopamine is made by certain cells in the brain. Dopamine carries messages that tell the body when and how to move, allowing for smooth, coordinated functions of the body's muscles. Parkinson's disease occurs when these cells either die or become damaged. When this happens there isn't enough dopamine to carry the messages and movement becomes very difficult.

Treatment methods involved implantation of stem cells that transform into dopamine-producing cells into the brains of Parkinson's disease patients. In one method, one man's own neural stem cells were extracted from his brain and stimulated to differentiate into dopamine-secreting cells *in vitro*. These cells are subsequently transplanted into the left side of the brain.

Another treatment method involved cultivating stem cells from another tissue so that they become dopamine-producing cells and injecting them into target sites of the patients' brain that need dopamine. These transplantations of cells have enabled patients to remain active.

(e) Describe how the stem cells are differentiated into dopamine-secreting cells *in vitro*. [3]

- **Grow on culture medium to ensure that they are capable of self renewal;;**
- **Induce differentiation by growing in a medium;;**
- **Which contains a mixture of hormones / protein growth factors and nutrients;;**
- **Hormones / protein growth factors produce desired changes in the cells to form dopamine-producing cells / Nutrients ensure cell survival;;**
- **During which gene expression is regulated / certain genes are switched on while other genes are switched off;;**

(f) "One treatment method involved cultivating stem cells from another tissue so that they become dopamine-producing cells..."

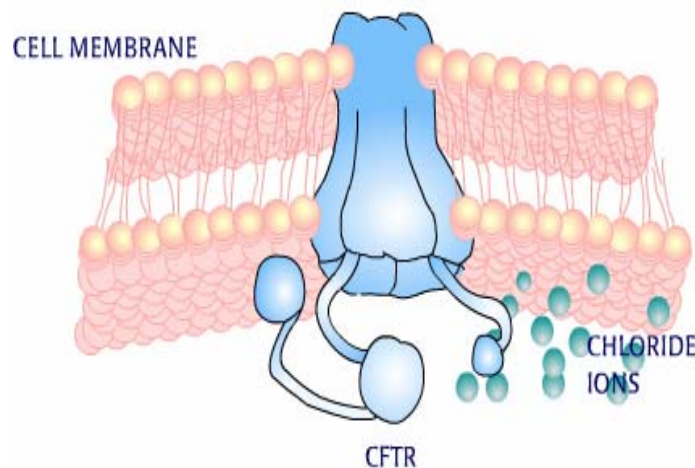
State the property of stem cells that allows for such transformation to occur. [1]

- **Plasticity/trans-differentiation;;**

[Total: 14 marks]

2. The first British trial of a possible gene therapy for cystic fibrosis (CF) began in 1993. Nine adult male patients with advanced lung disease were chosen. Six patients received gene therapy and three acted as a control.

The purpose of the trial was to see whether functioning ion channels could be manufactured by cells which have had the normal allele of the CF gene inserted into them. DNA coding for the normal allele was cloned and put into small spheres of phospholipids bilayers (liposomes) which were introduced into lung epithelial cells.



- (a) Describe the gene mutation and mode of inheritance of cystic fibrosis. [4]

- **Mutation in the cystic fibrosis transmembrane conductance regulator gene/CFTR gene;;**
- **Found on chromosome 7;;**
- **Deletion of 3 nucleotides CTT of the gene;;**
- **Loss of amino acid phenylalanine;;**
- **Autosomal recessive mode of inheritance;;**

- (b) Discuss why the liposome strategy is considered an example of gene therapy? [2]

- **It introduces normal/working/functional gene;;**
- **to epithelial cells of CF patients in order to treat the disorder by expression of the normal gene to restore normal function.;;**

- (c) Describe how a liposome introduces DNA into mammalian cells. [2]

- **Use of aerosol / sprays / inhalers to introduce liposomes into body;;**
- **Liposome fuses with the target cell membrane;;**
Normal genes move across membranes into cells

(d) Compare and contrast retroviral and adenoviral vectors used in gene therapy. [4]

Differences:

Retroviral vector	Adenoviral vector
Retroviral DNA can be integrated into the host cell genome;	Adenoviral DNA cannot be integrated into the host cell genome;
Retroviruses have low immunogenicity / less likely to stimulate immune response;	Adenoviruses have high immunogenicity / more likely to stimulate immune response;
Retroviruses can <u>only infect dividing cells</u> as they are unable to pass through the nuclear pores;	Adenoviruses <u>can infect both dividing and non-dividing cells</u> as they can enter the nucleus at any time via formation of a capsid-pore complex;
Retrovirus genome is a <u>ssRNA</u>;	Adenoviruses genome is a <u>dsDNA</u>;

Similarities:

- Both vectors have high efficiency of transfer.;;
- Both vectors need to bind to specific receptors before they can enter cells.;;

(e) Explain how the expression of the mutant cystic fibrosis alleles leads to the occurrence of bacterial infections in lungs. [3]

- CFTR gene provides instructions for making a channel that transports chloride ions into and out of cells;;
- The flow of chloride ions helps control the movement of water in tissues and regulates the fluid consistency of mucus / Mutations in the CFTR gene disrupt the normal function of this channel, preventing the usual flow of chloride ions and water;;
- Lung epithelium cells expressing mutant CF gene produce abnormally thick and sticky mucus which harbours bacteria and leads to bacterial infection;;

(f) Suggest why male patients were chosen for the British trial. [1]

- This is to avoid the risks of the allele being passed to the offspring;
- since virtually all men with CF are infertile.;

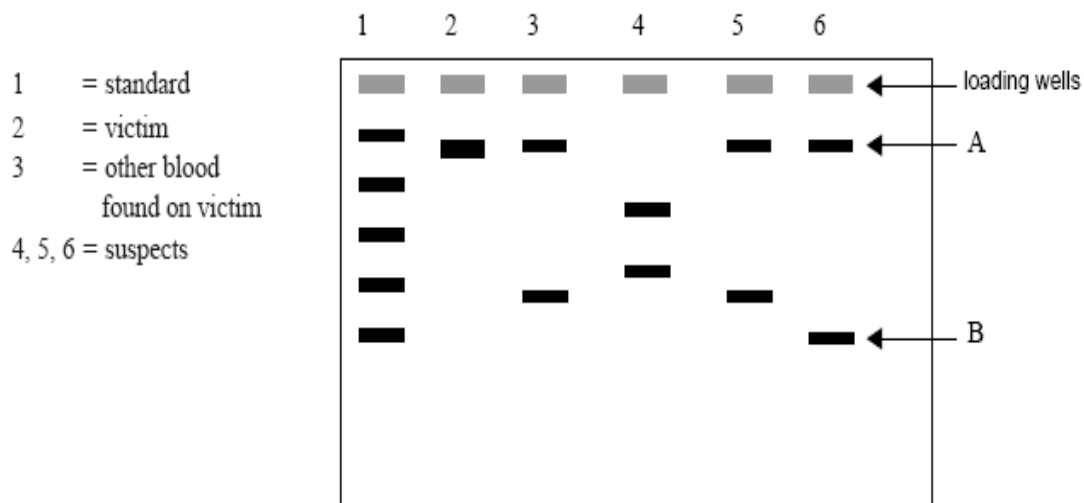
[Total: 16 marks]

3. There were three suspects in an assault case. A forensic scientist found blood, other than the victim's, at the site. DNA was extracted from five blood samples:
- The victim
 - The blood at the assault site (not the victim's)
 - The three suspects

Polymerase Chain Reaction (PCR) was carried out on the extracted DNA from the five blood samples.

One of the regions used in the forensic analysis was a short tandem repeat (STR) sequence of 4 bases, called **HUMTHO1**. This sequence, located on chromosome 11, has many alleles which differ from each other by the number of times the sequence **AATG** is repeated. It was this region of chromosome 11 which was amplified using PCR.

The amplified samples were loaded onto a gel and electrophoresis was performed to separate the fragments of DNA. A diagram of the gel is shown in the figure below.



(a) Why is there only one band in lane 2 but two bands in lanes 3, 4, 5 and 6? [2]

- There is only one band in lane 2 because **individual 2 is homozygous for the allele at chromosome 11;; (reject: homologous / blood is homozygous / chromosomes are homozygous)**
- **carrying two DNA fragments with the same number of repeats;;**

(while the others individuals are heterozygous for the allele on chromosome 11, carrying two DNA fragments with different number of repeats)

Reject: mutation resulted in the loss of RE sites, leading to uncut fragments → read story!

(b) How many different alleles at the **HUMTHO1** locus are represented on the gel in individuals 2, 3, 4, 5, 6? [1]

- There are 5 different alleles at the HUMTHO1 locus represented on the gel;;

(c) Which piece of DNA, A or B, has the greater number of the 4 base repeat sequence? Explain why. [2]

- DNA piece A has the greater number of the 4 base repeat sequence;;
- The greater the molecular weight / larger the DNA fragment of the sample the smaller distance the fragment will move from the loading well / nearer the fragment from the loading well;; (Note: to avoid terms 'heavier / lighter DNA')

(d) Which of the suspects appears to have committed the assault? Explain your choice. [2]

- Suspect whose sample is loaded in lane 5;;
- because the bands on the gel for suspect 5 match the sample of blood found on the victim, which was not the victim's blood (lane 3);;

(e) Explain briefly the basis behind Restriction Fragment Length Polymorphisms (RFLP). [2]

- Presence of different alleles of the same gene / short tandem repeats (STRs) / mutations;
- Differences in nucleotide sequences;
- Alter restriction sites present in the alleles / STRs;
- Thus different individuals having different alleles / different number of STRs will have different restriction fragments produced;

(f) State another application of RFLP analysis besides genetic fingerprinting. [1]

- Disease testing;;
- Genomic mapping;;

(Note: Genetic fingerprinting is associated with identity-testing, used for forensics, paternity tests, differentiate between different species etc)

(g) Explain why genes especially **housekeeping genes** are seldom used as markers for genetic fingerprinting. [2]

- Housekeeping genes encode functional polypeptides;
- Sequence of DNA must be conserved;
- due to its important biological function;
- Mutation rate lower than non- coding regions;
- Not able to differentiate between individuals;

The Human Genome Project (HGP) was one of the great feats of exploration in history, an inward voyage of discovery of the genome - of members of our species, *Homo sapiens*. The HGP was completed in April 2003. Now we can, for the first time, read nature's complete genetic blueprint for building a human being.

(h) Suggest how information from HGP can help scientists understand the role of genes in complex diseases like cancer. [3]

- HGP data comprises sequence of human genes and their chromosomal locations;
- helps scientists to determine the proteins encoded by genes;
- and predict/study their functions;
- Knowing the function of genes is first step in narrowing down which genes could be involved in the mechanism of a complex genetic disease / identify marker genes / genetic mutations associated with diseases;
- Knowing complete gene sequences helps scientist in studying how gene regulation/gene interactions may occur;
- and may lead to disease phenotypes;
- Knowing which genes encode for which proteins helps scientists correlate aberrant expression of proteins to their possible genetic causes;
- HGP data can be used for comparison of gene sequences between diseased and healthy individuals;
- HGP data allows for comparative genomics / comparative studies with genomes of other model organisms;
- useful to develop animal models of disease development;

[Total: 15 marks]

Section B

Answer the following question on the separate answer paper provided.

Your answer must be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections (a), (b) etc., as indicated in the question.

4. With reference to the analysis of DNA molecules,
a. Describe and explain the use of the Polymerase Chain Reaction (PCR). [10]

1. in vitro technique to amplify specific sequence of DNA in a short period of time;
2. 3 step process (denaturation, hybridization / annealing, extension / elongation);
3. carried out in thermal cyclers;

4. Denaturation;

5. heat reaction tube containing specific target dsDNA to ~ 94 – 96°C;

6. to break H- bonds;

7. separating dsDNA into ssDNA;

8. Hybridisation / annealing;

9. cool reaction mixture to ~ 60 – 65°C;

10. allow DNA (forward and reverse) primers to complementary base-pair with ssDNA template;

11. Extension/elongation;

12. raise temperature to ~ 70 – 75°C;

13. Taq / DNA polymerase binds to DNA template where primer has annealed;

14. add complementary nucleotides to 3' OH ends of primers;

15. catalyses phosphodiester bonds between adjacent nucleotides;

16. process repeats itself to produce multiple copies of target sequence;

17. highly sensitive, capable of amplifying sequences from minute amounts of target DNA; OWTTE
18. rapid and can generate many copies within a short period of time; OWTTE
19. robustness of the process and can be used on DNA that may not be as well-preserved; OWTTE
20. for (; for each application, up to max of 2 applications);;
 - DNA fingerprinting
 - studying human ancestry and ancient DNA
 - paternity testing
 - cloning genes for using in genetic manipulation
 - investigating gene expression
 - testing for detection bacteria/virus

b. Describe the process and explain the use of Gel Electrophoresis and Nucleic Acid Hybridisation. [10]

Gel Electrophoresis

1. DNA fragments are loaded in a well;
2. at one end of the agarose gel;
3. Electric field or voltage applied across the gel;
4. DNA being negatively charged;
5. will move towards the positive electrode/anode;
6. DNA ladder is added to calibrate the size of DNA; OWTTE
7. Distance travelled in a given time depends on the molecular weight / mass / size / length of the DNA;
8. with larger DNA fragments travel slower and smaller fragments traveling faster;
9. More resistance for the larger fragment to move through the pores of the gel;
10. Ethidium Bromide and UV light are used to visualize the DNA bands;
11. This allows for separation of a mixture of DNA fragments;;

[max 4m for description and 1 m for analysis of DNA]

Nuclei Acid Hybridisation

12. The separated DNA fragments in the gel are transferred to the nitrocellulose membrane by blotting;
13. During the process, the DNA fragments are denatured;
14. A single-stranded probe;
15. which is radioactively labelled; and
16. which is complementary to the target DNA sequence is added;
17. The probe hybridises to the fragment;
18. Excess probe is wash off;
19. The membrane is then exposed to a X-ray film / Autoradiography is carried out;
20. The region where probe bind to will give a dark band;
21. This allows for identification of different alleles / Restriction Fragment Length Polymorphisms;;
22. for genetic fingerprinting / paternity testing / disease testing / determination of evolutionary relationship;;

[Any 4 points - 4m for description and 1m for analysis of DNA]